KUSHAL AGRAWAL

Final Year Undergraduate at IIT Kanpur
Department of Biological Sciences and Bioengineering

KushalAgrawal123
Kushal Agrawal
kushalag23@iitk.ac.in

Degree/Certificate	Institute	Year	CPI/%
B.Tech	Indian Institute of Technology Kanpur	2023-2027	**8.2**/10
Class XII (CBSE)	Arcadian Public School, Mathura	2022	**96**%
Class X (CBSE)	Arcadian Public School, Mathura	2020	**96.8**%

COMPETITIVE PROGRAMMING ACHIEVEMENTS

- Achieved a maximum rating of **1619 (Expert)**, on **Codeforces** under the handle **KushalAgrawal**
- Achieved a maximum rating of **1817 (4 Stars)**, on **CodeChef** under the handle **agrkushal123**
- Solved **700+ problems**, across **Codeforces**, **LeetCode**, **CSES**, **CodeChef**, **GeeksforGeeks** and **InterviewBit**

WORK EXPERIENCE

Center for Developing Intelligent Systems *May'26-Jul'26*

Software Engineer Intern

- Developed a production **healthcare analytics platform** for ECHS, processing **26M+ claims** worth **₹37,800+ Cr** to support fraud detection and investigation workflows
- Built and deployed large-scale **AI-powered CCTV exam surveillance system** supporting **100,000+ camera feeds** for NEET, enabling real-time monitoring and integrity detection
- Developed production-grade backend systems using **Node.js**, **Express**, **React** and **MySQL**, building REST APIs, authentication, dashboards, optimization and reports
- Built ECHS fraud analytics pipelines using **Python**, **pandas** and **SQL**, developing ETL workflows, anomaly detection, provider benchmarking and cost analysis for suspicious claims
- Architected computer vision pipelines using **vision-language models**, **YOLO detection** and **multi-object tracking** to classify scenes and detect exam-integrity events
- Engineered scalable production infrastructure with **Docker services**, **GPU inference workers**, optimized queries and deployment workflows for reliable real-time processing
- Developed multi-signal investigation frameworks using statistical analysis, claim insights and cross-source correlation to identify high-risk entities and generate **audit-ready reports**

CarbonTatva AI Pvt. Ltd. *Mar'26-Apr'26*

Machine Learning Intern

- Developed electricity price forecasting solutions for **GDAM (Green Day-Ahead Market)**, addressing energy market prediction challenges through machine learning approaches
- Processed **433K+ time-series records** spanning 4 years to build forecasting datasets and enable **24 hours** and **36 hours ahead** energy market predictions
- Developed ML pipelines integrating market and weather data, performing preprocessing, feature engineering, and temporal modeling for energy forecasting tasks
- Implemented and evaluated ML/DL models including **XGBoost**, **LightGBM**, **CNN-LSTM**, **N-BEATS** and **Temporal Fusion Transformer** to improve forecasting accuracy
- Analyzed historical trends and distribution shifts to enhance model robustness and deliver forecasting solutions for climate-tech energy analytics applications

TECHNICAL SKILLS

- **Languages:** C++, Python, JavaScript, Java, SQL
- **Libraries and Frameworks:** TensorFlow, Pytorch, Matplotlib, Numpy, Pandas, Keras, Scikit Learn, Seaborn, OpenCV
- **Development:** HTML, CSS, Tailwind, Node.js, Express, React, Springboot, Kafka, FastAPI
- **Others:** Git, GitHub, MS Excel, Tableau, Linux

PROJECTS

Novel Natural Adversarial Attacks and Defense *Jan'26-Apr'26*

Mentor: Prof. Koteswar Rao, EE655 Course Project

- Trained a **CIFAR-10 ResNet-18** to **95.5% clean accuracy** in PyTorch and benchmarked robustness under realistic transforms
- Implemented **4 differentiable natural adversarial attacks** — spatial, fog, snow and color — via gradient-based optimization
- Designed a **novel composite attack** jointly optimizing multiple perturbations, dropping accuracy **95.5% to 47.1%** (50.9% ASR)
- Built **LPIPS-based perceptual metrics** and cross-attack analysis to compare attack strength and expose model vulnerabilities
- Developed mixed adversarial fine-tuning and a Streamlit demo, recovering **+14.7 robustness points** with no clean-accuracy loss

DICE-FER *May'25-Jul'25*

Mentor: Prof. Nishchal K. Verma, EE656 Course Project

- Developed **facial expression recognition** using dual-encoder **ResNet-18** to disentangle identity/expression features
- Designed an information-theoretic disentanglement framework using **MINE**, Donsker-Varadhan bounds and MI optimization
- Built deep learning pipeline for CK+, Oulu-CASIA, RAF-DB and AffectNet with augmentation, checkpointing and TensorBoard
- Achieved **99.5% CK+**, **95.5% Oulu-CASIA** and **85.5% RAF-DB** accuracy with MIG-based representation evaluation

CollabOS AI: Collaboration Platform *Jan'26-Apr'26*

Self Project

- Built **multi-tenant collaboration platform** using **Spring Boot**, **React**, **PostgreSQL** and **Redis** with RBAC and workspaces
- Implemented **WebSocket-based** real-time collaboration and messaging via **Kafka-driven workflows**
- Built a **RAG-based** document pipeline using **FastAPI**, **LangChain** and **pgvector** for semantic search
- Developed **LangGraph AI agents** to query project data, identify risks and automate workflow assistance
- Containerized the platform with **Docker**, integrating modular AI services with configurable LLMs and testing

TinyKV: Distributed Storage Engine *Jan'26-Apr'26*

Self Project

- Built **Redis-inspired key-value engine** in **C++17** with custom TCP protocol, command processing and in-memory storage
- Designed **multithreaded server architecture** using sockets, synchronization, hash tables, LRU caching, and TTL expiration
- Implemented **crash-safe persistence** using append-only logs and snapshot recovery for reliable data restoration
- Developed distributed storage features with **primary-replica replication** and **consistent-hash sharding** for scalability
- Added **fault-tolerant capabilities** including failover handling, recovery, and automated testing for reliability

Multimodal sentiment analysis *Dec'24- Feb'25*

Google Developer's Group, IIT Kanpur

- Developed **NLP pipelines** for sentiment analysis using preprocessing, lemmatization, and **transformer features**
- Fine-tuned **DistilBERT** using **HuggingFace Trainer** on **50K IMDB reviews**, achieving **93.58% accuracy** with GPU training
- Built a **multimodal meme pipeline** combining text/image features via transformer preprocessing, PCA and neural networks
- Implemented ML workflows using **PyTorch**, **TensorFlow/Keras** and **scikit-learn** with tuning, evaluation and persistence

RELEVANT COURSES

AI, ML, DL and Its Applications
Applied Probability and Statistics
Computer Vision and Deep Learning
Single Variable Calculus
Fundamentals of Computing
Several Variable Calculus
Linear Algebra
ODE